

CSCI 2540: Advanced Probabilistic Methods in Computer Science

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Why Probabilistic Methods?

- Randomized algorithms (hashing, quicksort, open key cryptography, finance, weather)
- Probabilistic analysis of algorithms (average case analysis, smoothed analysis)
- Modeling data (statistical ML)

Course Topics:

1. Basic randomized algorithms and probabilistic analysis
2. Large deviation bounds (Chernoff and Hoeffding bounds)
3. The probabilistic method
4. Martingale (in discrete space)
5. Sample complexity and statistical learning theory
6. Monte Carlo methods
7. Converge of Monte Carlo Markov Chains

Example (Verifying Matrix Multiplication): Suppose you have three $n \times n$ matrices A, B, C in a boolean field. We want to verify $AB = C$.

A naive matrix multiplication algorithm takes $\Theta(n^3)$ time ($\Theta(n^{2.37})$ at the absolute best). But we would hope verification can be done faster.

Algorithm 1: Freivalds' Randomized algorithm in $\Theta(n^2)$

- 1 Choose a random vector $r \in \{0, 1\}^n$ uniformly at random.
 - 2 Compute Br and $A(Br)$
 - 3 Compute Cr
 - 4 If $A(Br) \neq Cr$, return " $AB \neq C$ ". Otherwise, return " $AB = C$ ".
-

Theorem: If $AB \neq C$ and r is chosen uniformly at random from $\{0, 1\}^n$, then

$$\mathbb{P}(ABr = Cr) \leq \frac{1}{2}$$

Proof:

Take the case $AB \neq C$. Let $D = AB - C \neq 0$. Then

$$ABr = Cr \implies Dr = 0$$

Since $D \neq 0$, there exists some $d_{ij} \in D \neq 0$. By definition of matrix multiplication, we have

$$\sum_{j=1}^n d_{1j} r_j = 0 \iff r_1 = -\frac{1}{d_{11}} \sum_{j=2}^n d_{1j} r_j$$

(assuming WLOG $d_{11} \neq 0$).

Letting $a_1, \dots, a_n$ being the values chosen at random from the set of variables $\{r_1, \dots, r_n\}$, we have

$$\mathbb{P}\left(r_1 = -\frac{1}{d_{11}} \sum_{j=2}^n d_{1j} a_j\right) \leq \frac{1}{2}$$

Hence, if $AB = C$ the algorithm is always correct. If $AB \neq C$ the algorithm is correct with probability at least $1/2$. We can repeat the algorithm k times to reduce the error probability to $1/2^k$.

Probability space: $(\Omega, \mathcal{F}, \mathbb{P})$ for Ω a sample space of outcomes, $\mathcal{F}$ a σ -algebra of allowable events (subsets of Ω), and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ a probability measure.

In a **discrete probability space**, $\omega \in \Omega$ is a simple event and $\mathcal{F} = 2^\Omega$.

Probability function: a function $\mathbb{P} : \mathcal{F} \rightarrow \mathbb{R}$ that satisfies

1. For all $E \subseteq \mathcal{F}$, $0 \leq \mathbb{P}(E) \leq 1$
2. $\mathbb{P}(\Omega) = 1$
3. For any countable collection of pairwise mutually disjoint events $E_1, E_2, \dots \in \mathcal{F}$, we have

$$\mathbb{P}\left(\bigcup_{i \geq 1} E_i\right) = \sum_{i \geq 1} \mathbb{P}(E_i)$$

In a discrete space, each *atomic event* has a finite probability and the probability of an event E is given by the sum of the probabilities of the atomic events in E

Let's return to Freivald's algorithm,

Lemma: Choosing $r = (r_1, \dots, r_n) \in \{0, 1\}^n$ uniformly at random is equivalent to choosing each r_i independently and uniformly at random from $\{0, 1\}$

Hence a more formal version of the earlier proof proceeds:

$$\mathbb{P}(ABr = Cr) = \sum_{x_2, \dots, x_n} \mathbb{P}((ABr = Cr) \cap ((r_2, \dots, r_n) = (x_2, \dots, x_n))) \quad (1)$$

$$\leq \sum_{x_2, \dots, x_n} \mathbb{P}\left(\left[r_1 = -\frac{1}{d_{11}} \sum_{j=2}^n d_{1j} x_j\right] \cap ((r_2, \dots, r_n) = (x_2, \dots, x_n))\right) \quad (2)$$

$$= \sum_{x_2, \dots, x_n} \mathbb{P}\left(r_1 = -\frac{1}{d_{11}} \sum_{j=2}^n d_{1j} x_j\right) \mathbb{P}((r_2, \dots, r_n) = (x_2, \dots, x_n)) \quad (3)$$

$$\leq \sum_{x_2, \dots, x_n} \frac{1}{2} \mathbb{P}((r_2, \dots, r_n) = (x_2, \dots, x_n)) \quad (4)$$

$$= \frac{1}{2} \quad (5)$$

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0.2.1 Deciding Polynomial Identity

Problem: Verify that $P(x) \equiv Q(x)$ for two polynomials P, Q , i.e. $H(x) = P(x) - Q(x) \equiv 0$.

Example: Check if

$$(x+1)(x-2)(x+3)(x-4)(x+5)(x-6) \equiv x^6 - 7x^3 + 25$$

Notational remark: we use $\equiv$ to denote polynomial identity and $=$ for numerical equality.

Algorithm 2: Randomized Polynomial Identity

- 1 Choose $r \in [1, \dots, m]$;
 - 2 If $H(r) = 0$ output “ $P \equiv Q$ ” else output “ $P \not\equiv Q$ ”
-

Theorem: Assume that $H(x) \not\equiv 0$ and $\deg(H) \leq d$. Then if r is chosen uniformly at random from $\{1, \dots, m\}$, we have

$$\mathbb{P}(H(r) = 0) \leq \frac{d}{m}$$

Proof: A polynomial of degree d has at most d roots. Hence, if $H(x) \not\equiv 0$, then there are at most d values of r such that $H(r) = 0$. Since r is chosen uniformly at random from $\{1, \dots, m\}$, we have

$$\mathbb{P}(H(r) = 0) \leq \frac{d}{m}$$

What about multivariate polynomials?

Example:

$$H(x, y, z, w) = x^5 y^3 (z^3 - w^4) + x^3 (z^5 - w^3) (x^2 y^4 - w^2) \stackrel{?}{=} 0$$

Define the degree of a multivariate polynomial $H(x_1, \dots, x_n)$ as the maximum sum of the degrees in each of its monomials. For example, the degree of $H(x, y, z, w)$ above is $3 + 5 + 2 + 4 = 14$.

For this setting, no efficient deterministic algorithm is known. A polynomial with n variables and degree d can have up to $\binom{n+d}{d}$ monomials, which is exponential in n

Theorem (Schwartz-Zippel): Assume that $H(x_1, \dots, x_n) \not\equiv 0$ and $\deg(H) \leq d$. Then if $r_1, \dots, r_n$ are chosen independently and uniformly at random from a set S (equivalently, sampling $(r_1, \dots, r_n)$ from S^n), then

$$\mathbb{P}(H(r_1, \dots, r_n) = 0) \leq \frac{d}{|S|}$$

Proof: By induction on n .

For $n = 1$, H has at most d roots and by the theorem above,

$$\mathbb{P}(H(r_1) = 0) \leq \frac{d}{|S|}$$

For $n > 1$, H can be written as a polynomial in x_n with coefficients that are polynomials in $x_1, \dots, x_{n-1}$:

$$H(x_1, \dots, x_n) = \sum_{i \geq 0} x_n^i f_i(x_1, \dots, x_{n-1})$$

Let $k \leq d$ be the largest index for which $f_k \not\equiv 0$. Then $\deg(f_k) \leq d - k$ and by the induction hypothesis,

$$\mathbb{P}(f_k(r_1, \dots, r_{n-1}) = 0) \leq \frac{d - k}{|S|}$$

If $f_k(r_1, \dots, r_{n-1}) \neq 0$, then $H(r_1, \dots, r_{n-1}, x_n)$ is a nonzero univariate polynomial in x_n of degree k . Hence by the univariate case,

$$\mathbb{P}[H(r_1, \dots, r_n) = 0 \mid f_k(r_1, \dots, r_{n-1}) \neq 0] \leq \frac{k}{|S|}$$

Then since

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) \\ &= \mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c) \\ &\leq \mathbb{P}(A \mid B) + \mathbb{P}(B^c)\end{aligned}$$

we have

$$\begin{aligned}\mathbb{P}(H(r_1, \dots, r_n) = 0) &= \mathbb{P}[H(r_1, \dots, r_n) = 0 \mid f_k(r_1, \dots, r_{n-1} \neq 0)] + \mathbb{P}[f_k(r_1, \dots, r_{n-1}) = 0] \\ &\leq \frac{k}{|S|} + \frac{d-k}{|S|} \\ &= \frac{d}{|S|}\end{aligned}$$

Recall the definition of conditional probability.

Conditional Probability:

$$\mathbb{P}(E_1 \mid E_2) = \frac{\mathbb{P}(E_1 \cap E_2)}{\mathbb{P}(E_2)}$$

where we can think of conditioning on E_2 as restricting the sample space to E_2 . Hence, we have

$$\mathbb{P}(E_n) = \sum_{i=1}^n \mathbb{P}(E_n \mid E_i) \mathbb{P}(E_i)$$

This exposes a general method for the course: “Suppose some nice condition holds and analyze the algorithm conditioned on that event. Then bound the probability of that event not holding.”

This theorem has a **one-sided error**: if $H \equiv 0$, the algorithm is always correct. If $H \not\equiv 0$, the algorithm is correct with probability at least $1 - d/|S|$. We can repeat the algorithm ℓ independent times to reduce the error probability to $(d/|S|)^\ell$.

Independent Events: $E \perp F$ iff

$$\mathbb{P}(E \cap F) = \mathbb{P}(E) \cdot \mathbb{P}(F)$$

More generally, events $E_1, \dots, E_n$ are **mutually independent** iff for any subset $I \subseteq \{1, \dots, n\}$,

$$\mathbb{P}\left(\bigcap_{i \in I} E_i\right) = \prod_{i \in I} \mathbb{P}(E_i)$$

In particular, if $E \perp F$, then $\mathbb{P}(E \mid F) = \mathbb{P}(E)$.

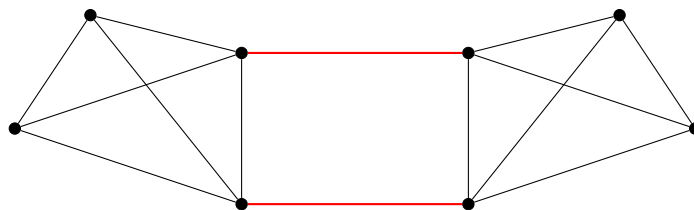
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Min-cut set: a minimum set of edges that disconnects a graph.

Trivially, we can iterate over all nodes and cut all edges for the node with the minimum degree. This gives an upper bound on the size of the min-cut:

$$\text{min-cut} \leq \min_{v \in V} \deg(v)$$

Consider, however, this graph:

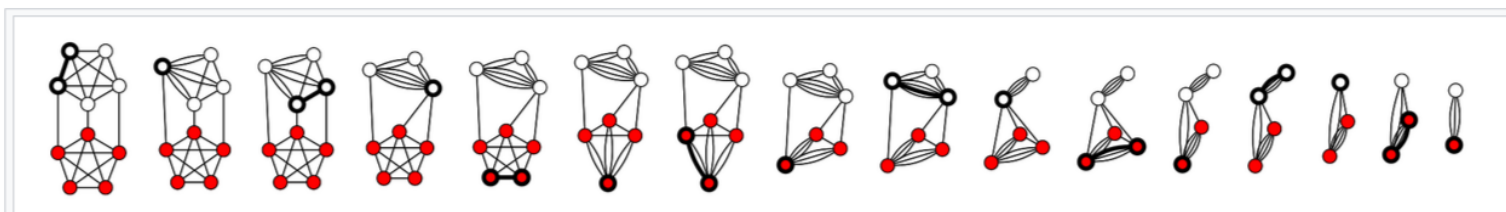


Every node has degree 4, but the min-cut is 2.

As we will see, this is a deep problem with many applications. This problem happens to have a deterministic polynomial time algorithm related to max-flow but it is complicated and involves an $O(|V|^2)$ algorithm run between every pair of nodes ($O(\binom{|V|}{2})$ pairs).

Algorithm 3: Randomized Min-cut Algorithm

- 1 **for** $i=1:n-2$ **do**
 - 2 Pick an edge uniformly at random;
 - 3 Contract the two vertices connected by that edge, eliminate all edges connecting the two vertices;
 - 4 **Output** the set of edges connecting the two remaining vertices;
-



Successful run of Karger's algorithm on a 10-vertex graph. The minimum cut has size 3.

Notice that this algorithm fails precisely when we contract an edge in the min-cut. Following the strategy we saw last time, we would like to find the condition that makes the algorithm fail deterministically and then bound the probability of that condition holding.

Theorem:

1. The algorithm outputs a min-cut edge-set with probability

$$\geq \frac{2}{n(n-1)}$$

2. The smallest set output in $O(n^2 \log n)$ iterations of the algorithm gives a correct answer with probability $1 - 1/n^2$.

Proof: Assume the graph has a min-cut set of k edges. We compute the probability of finding one particular such set C .

Trivially,

$$\mathbb{P}(\text{alg. returns any min-cut}) \geq \mathbb{P}(\text{alg. returns min-cut } C)$$

Lemma: If no edge of C was contracted, the algorithm outputs C .

First, notice that if no edge of C was contracted, no edge of C was eliminated. To see this, let X and Y be the two sets of vertices connected by edges in C . If no edge of C was contracted, then no vertex in X was contracted with a vertex in Y . Hence, all edges in C remain in the graph and are output by the algorithm.

As a corollary, if the algorithm terminates before contracting any edge in C , it gives the correct answer.

Finally, to show the one-sided error property, notice that vertex contraction does not reduce the size of the min-cut set but does reduce the size of the original graph. Intuitively, contracting an edge *reduces the size of the sample space*

Lemma:

$$\mathbb{P}(\text{no edge in } C \text{ is contracted}) \geq \frac{2}{n(n-1)}$$

Let E_i be the event “no edge contracted in iteration i is in C ” so

$$F_i = \bigcap_{j=1}^i E_j = \text{“no edge of } C \text{ is contracted in the first } i \text{ iterations”}$$

Since the min-cut set has k edges, all vertices have degree $\geq k$ (else the min-cut set would be smaller). Hence, the number of edges in the graph $\geq nk/2$.

$$\mathbb{P}(E_1) = \mathbb{P}(F_1) = 1 - \frac{2k}{nk} = 1 - \frac{2}{n}$$

Conditioning on E_1 , after the first contraction, we have $n-1$ vertices and $k(n-1)/2$ edges but still k edges in the min-cut set. By analogous argument,

$$\begin{aligned} \mathbb{P}(E_2) &= \mathbb{P}(E_2 \mid E_1) \cdot \mathbb{P}(E_1) + \mathbb{P}(E_2 \mid \bar{E}_1) \cdot \mathbb{P}(\bar{E}_1) \\ \mathbb{P}(E_2 \mid E_1) &\geq 1 - \frac{k}{|G_1|} \quad (\text{where } |G_1| = \# \text{ edges after iter 1}) \\ &\geq 1 - \frac{k}{(n-1)k/2} \\ &= 1 - \frac{2}{n-1} \end{aligned}$$

Of course, since $\mathbb{P}(E_1) = \mathbb{P}(F_1)$,

$$\mathbb{P}(E_2 \mid F_1) = \mathbb{P}(E_2 \mid E_1) \geq 1 - \frac{k}{k(n-1)/2} \geq 1 - \frac{2}{n-1}$$

Inductively,

$$\begin{aligned} \mathbb{P}(E_i \mid F_{i-1}) &= \mathbb{P}\left(E_i \mid \bigcap_{j=1}^{i-1} E_j\right) \\ &\geq 1 - \frac{k}{k(n-i+1)/2} \\ &= 1 - \frac{2}{n-i+1} \end{aligned}$$

How do we compute F_{n-1} given a sequence of $\mathbb{P}(E_i \mid F_{i-1})$? By chain rule:

$$\begin{aligned} \mathbb{P}(F_n) &= \mathbb{P}\left(\bigcap_{j=1}^n E_j\right) \\ &= \mathbb{P}(E_{n-1} \mid F_{n-1}) \cdot \mathbb{P}(F_{n-1}) \\ &= \mathbb{P}(E_n \mid F_{n-1}) \cdot \mathbb{P}(E_{n-1} \mid F_{n-2}) \cdots \mathbb{P}(E_2 \mid F_1) \cdot \mathbb{P}(E_1) \end{aligned}$$

We are interested in $\mathbb{P}(F_{n-2})$ since the algorithm terminates after $n - 2$ iterations. Simply plugging in,

$$\begin{aligned}
 \mathbb{P}(F_{n-2}) &= \mathbb{P}(F_1) \cdot \prod_{j=2}^{n-2} \mathbb{P}(E_j \mid F_{j-1}) \\
 &\geq \prod_{i=1}^{n-2} \left(1 - \frac{2}{n-i+1}\right) \\
 &= \prod_{i=1}^{n-2} \frac{n-i-1}{n-i+1} \\
 &= \frac{n-2}{n} \cdot \frac{n-3}{n-1} \cdot \frac{n-4}{n-2} \cdots \frac{2}{n-1} \\
 &= \frac{2}{n(n-1)} \quad \blacksquare
 \end{aligned}$$

Proof of Part 2: Assume that we run the algorithm $n(n-1) \log n$ times and output the minimum cut set found across the iterations. We claim the probability of not finding a min-cut set is at most $1/n^2$.

From the lemma proving the one-sided error, the output cannot be smaller than the min-cut set. Hence, the probability that C is *not* the output of any of the $n(n-1) \log n$ independent iterations is

$$\leq \left(1 - \frac{2}{n(n-1)}\right)^{n(n-1) \log n} \leq e^{-2 \log n} = \frac{1}{n^2}$$

Via Taylor Series,

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$$

So for $x < 1$,

$$1 - x \leq e^{-x}$$

Once again, we can repeat this algorithm to reduce the error because it has **one-sided error** (the algorithm is completely correct for one specific outcome while bounding the error probability for the other outcome)

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Algorithm 4: Quicksort

- 1 **Input:** An array S
- 2 **Output:** The array S in sorted order.
- 3 Choose an element $y \in S$
- 4 Compare all elements of S to y . Let

$$\begin{aligned}
 S_1 &= \{x \in S - \{y\} \mid x \leq y\} \\
 S_2 &= \{x \in S - \{y\} \mid x > y\}
 \end{aligned}$$

- 5 Return $\text{Quicksort}(S_1) \cup \{y\} \cup \text{Quicksort}(S_2)$
-

What can we say about the run time of this algorithm? Well, it depends on what it means to “choose an element $y \in S$ ”.

In the *randomized* version of the algorithm, we choose y uniformly at random from S .

But if we were to choose y deterministically (say splitting the array in half), we could end up with a worst case run time of $O(n^2)$ or else do well and get $O(n \log n)$ run time.

Let $T(n)$ be the number of comparisons made by the algorithm on an array of size n .

Here, $T(n)$ is a *random variable*.

Random Variable: X is a random variable if it is a function $X : \Omega \rightarrow \mathbb{R}$ from the sample space Ω to the real numbers.

Random Vector: $X^d : \Omega \rightarrow \mathbb{R}^d$

Discrete random variable: a random variable that takes on only a finite or countably infinite number of values. In this case, we use notation

$$\mathbb{P}(X = a) = \mathbb{P}(\{s \in \Omega : X(s) = a\}) = \sum_{s \in \Omega : X(s) = a} \mathbb{P}(s)$$

Viewing random variables as functions of events, we can generalize the definition of independence to random variables:

Independent Random Variables: $X \perp Y$ iff for all $x, y \in \Omega$,

$$\mathbb{P}(X = x \wedge Y = y) = \mathbb{P}(X = x) \cdot \mathbb{P}(Y = y)$$

Similarly, $X_1, \dots, X_n$ are **mutually independent** iff for all $a_1, \dots, a_n \in \mathbb{R}$,

$$\mathbb{P}(X_1 = a_1 \wedge \dots \wedge X_n = a_n) = \prod_{i=1}^n \mathbb{P}(X_i = a_i)$$

Median: the median(s) of a random variable is a value m such that

$$\begin{cases} \mathbb{P}(x < m) \leq \frac{1}{2} \\ \mathbb{P}(X > m) < \frac{1}{2} \end{cases}$$

for a symmetric distribution, the median is the same as the mean.

Finally, we can define the expectation of a random variable:

$$\mathbb{E}[X] = \sum_{i \in \text{range}(X)} i \cdot \mathbb{P}(X = i)$$

where $\mathbb{E}[X]$ exists iff $\sum_i |i| \mathbb{P}(X = i) < \infty$ (i.e. the sum converges uniformly).

Theorem: The expected number of steps in the randomized version of quicksort is

$$\mathbb{E}[T(n)] = O(n \log n)$$

Proof: Let $s_1, \dots, s_n$ be the sorted elements of S . For $i = 1 : n$ and $j > i$, define

$$X_{i,j} = \begin{cases} 1 & \text{if } s_i, s_j \text{ are compared in the algorithm} \\ 0 & \text{otherwise} \end{cases}$$

(for example, after the first partition elements of S_1 are never compared to elements of S_2). Then

$$T(n) = \sum_{i=1}^n \sum_{j>i} X_{i,j}$$

By linearity of expectation,

$$\mathbb{E}[T(n)] = \sum_{i=1}^n \sum_{j>i} \mathbb{E}[X_{i,j}]$$

Since $X_{i,j}$ is a Bernoulli random variable,

$$\mathbb{E}[X_{i,j}] = 0 \cdot \mathbb{P}(X_{i,j} = 0) + 1 \cdot \mathbb{P}(X_{i,j} = 1) = \mathbb{P}(X_{i,j} = 1)$$

Notice that s_i and s_j are compared if and only if one of them is chosen as the pivot before any of the $j-i-1$ elements between them is chosen.

Since elements are chosen uniformly at random, $X_{i,j} = 1$ iff the first pivot chosen in the set is $\{s_i, s_{i+1}, \dots, s_j\}$ is either s_i or s_j .

$$\mathbb{E}[X_{i,j}] = \frac{2}{j-i+1}$$

Substituting back into the expectation of $T(n)$, we have

$$\begin{aligned} \mathbb{E}[T(n)] &= \sum_{i=1}^n \sum_{j>i} \frac{2}{j-i+1} \\ &\leq n \sum_{k=1}^n \frac{2}{k} \\ &\leq 2nH_n \\ &= 2n \log n + O(n) \end{aligned}$$

where H_n is the harmonic series

$$H_n = \sum_{i=1}^n \frac{1}{i} \approx \int_1^n \frac{1}{x} dx = \log n$$

Notice in particular that this analysis did not require any knowledge of the distribution of the input array S .

Exercise: Prove

1. $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$
2. for $c \in \mathbb{R}$ and X an RV, $\mathbb{E}[cX] = c\mathbb{E}[X]$

Exercise: Prove $\log n \leq H_n \leq \log n + 1$

Algorithm 5: Deterministic Quicksort

- 1 **Input:** An array S
- 2 **Output:** The array S in sorted order.
- 3 Let y be the first element in S
- 4 Compare all elements of S to y . Let

$$S_1 = \{x \in S - \{y\} \mid x \leq y\}$$

$$S_2 = \{x \in S - \{y\} \mid x > y\}$$

- 5 Return $\text{DetQuicksort}(S_1) \cup \{y\} \cup \text{DetQuicksort}(S_2)$
-

Theorem: the expected run time of deterministic Quicksort on a random permutation of S is $O(n \log n)$

Proof: Set $X_{i,j}$ as before. If all permutations have equal probability, then permutations of $S_i, \dots, S_j$ have equal probability.

Again,

$$\mathbb{P}(X_{i,j} = 1) = \frac{2}{j - i + 1}$$

and

$$\mathbb{E}[T(n)] = \sum_{i=1}^n \sum_{j>i} \frac{2}{j - i + 1} = O(n \log n)$$

This opens the way for two classifications of uses of randomness in algorithms:

1. **Randomized algorithm:** algorithms where the analysis is true for all inputs, repeated runs are independent, and the sample space is the space of random choices made by the algorithm.
2. **Probabilistic Analysis of algorithms:** The sample space is the space of all possible inputs; If the algorithm is deterministic repeated runs give the same output.

In particular we will be interested in:

1. **Monte Carlo Algorithm:** a randomized algorithm that may produce an incorrect solution.
2. **Las Vegas algorithm:** a randomized algorithm that always produces the correct output.

For decision problems: A *one-side error Monte Carlo algorithm* errs only one one possible output, otherwise it is a two-side error algorithm.

In both types of algorithms the run-time is a random variable.

The Random Allocations Paradigm: set of algorithms where we randomly allocate m balls into n bins.

- Used for load balancing, hashing, routing, distributed computing, and more

Algorithm 6: The Coupon Collector Problem

- 1 **Input:** n bins
 - 2 **Output:** The number of balls placed until every bin has at least one ball.
 - 3 Place balls independently and uniformly at random into n bins.
 - 4 Continue until every bin has at least one ball. Let X be the number of balls placed.
-

Theorem:

1. $\mathbb{E}[X] = nH_n = n \log n + \Theta(n)$
2. $\mathbb{P}(X \geq 2H_n) = \Theta(1/n)$

Let X_i be the number of balls placed when there were $i - 1$ non-empty bins. Then

$$X = \sum_{i=1}^n X_i$$

so X_i is the number of trials until we hit an empty box, that is: X_i is a geometric random variable with success probability

$$p_i = 1 - \frac{i-1}{n}$$

Proof:

Geometric random variable: X is a geometric random variable with success probability p on $n = 1, 2, \dots$

$$\mathbb{P}(X = n) = (1 - p)^{n-1}p$$

(Commonly used for the number of trials until the first success in a sequence of independent Bernoulli trials.)

Lemma (Memoryless Distribution): For a geometric random variable X with success probability p ,

$$\mathbb{P}(X = n + k \mid X > k) = \mathbb{P}(X = n)$$

Proof:

$$\begin{aligned}
 \mathbb{P}(X = n + k \mid X > k) &= \frac{\mathbb{P}(X = n + k \wedge X > k)}{\mathbb{P}(X > k)} \\
 &= \frac{\mathbb{P}(X = n + k)}{\mathbb{P}(X > k)} \\
 &= \frac{(1 - p)^{n+k-1}p}{\sum_{i=k}^{\infty} (1 - p)^i p} \\
 &= \frac{(1 - p)^{n+k-1}p}{(1 - p)^k} \\
 &= (1 - p)^{n-1}p \\
 &= \mathbb{P}(X = n)
 \end{aligned}$$

Note that the geometric distribution is the only discrete memoryless distribution (and the exponential distribution is the only continuous memoryless distribution).

Exercise: Prove that the expectation of $X \sim G(p)$ is

$$\mathbb{E}[X] = \sum_{i \geq 1} i(1 - p)^{i-1}p = \frac{1}{p}$$

Conditional Expectations:

$$\mathbb{E}[Y \mid Z = z] = \sum_y yP(Y = y \mid Z = z)$$

Lemma: For any random variables X, Y

$$\mathbb{E}[X] = \mathbb{E}_Y[E_X[X | Y]] = \sum_y \mathbb{P}(Y = y) \mathbb{E}[X | Y = y]$$

Proof:

$$\begin{aligned} \mathbb{E}_Y[E_X[X | Y]] &= \sum_y \mathbb{P}(Y = y) \mathbb{E}[X | Y = y] \\ &= \sum_y \mathbb{P}(Y = y) \sum_x x P(X = x | Y = y) \\ &= \sum_y \mathbb{P}(Y = y) \sum_x x \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} \\ &= \sum_y \sum_x x P(X = x, Y = y) \\ &= \sum_x x P(X = x) \\ &= E[X] \end{aligned}$$

This gives us an easier way to compute the expectation of $X \sim G(p)$:

Proposition: Let $X \sim G(p)$ be a geometric random variable with success probability p . Then

$$\mathbb{E}[X] = \frac{1}{p}$$

Proof:

Let $Y = 1$ if the first trial is a success and $Y = 0$ otherwise. Then

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{P}(Y = 0) \mathbb{E}[X | Y = 0] + \mathbb{P}(Y = 1) \mathbb{E}[X | Y = 1] \\ &= (1 - p) \mathbb{E}[X | Y = 0] + p \mathbb{E}[X | Y = 1] \end{aligned}$$

Trivially, $\mathbb{E}[X | Y = 1] = 1$. In the case $Y = 0$, let Z be the number of trials until the first success after the first trial.

By the memoryless property, $Z \sim G(p)$ and $\mathbb{E}[Z] = \mathbb{E}[X]$. Hence,

$$\mathbb{E}[X] = (1 - p) \mathbb{E}[Z + 1] + p = (1 - p) \mathbb{E}[Z] + 1 \implies \mathbb{E}[X] = \frac{1}{p}$$

0.5 Sept 24

Recall: In the *Coupon Collector Problem*, we place balls independently and uniformly at random into n bins. We continue until every bin has at least one ball. How many balls do we expect to place?

Last time, we showed that

$$X = \sum_{i=1}^n X_i$$

where X_i is a geometric random variable with success probability $p_i = 1 - \frac{i-1}{n}$ representing the number of trials till we hit a new bin when there were already $i - 1$ non-empty bins.

Lemma: Let X be a discrete random variable that takes on only non-negative integer values. Then

$$\mathbb{E}[X] = \sum_{i=1}^{\infty} \mathbb{P}(X \geq i)$$

Proof:

$$\begin{aligned} \sum_{i=1}^{\infty} \mathbb{P}(X \geq i) &= \sum_{i=1}^{\infty} \sum_{j=i}^{\infty} \mathbb{P}(X = j) \\ &= \sum_{j=1}^{\infty} \sum_{i=1}^j \mathbb{P}(X = j) \\ &= \sum_{j=1}^{\infty} j \mathbb{P}(X = j) \\ &= \mathbb{E}[X] \end{aligned}$$

By the lemma and the fact that $X \sim G(p)$,

$$\begin{aligned} \mathbb{P}(X \geq i) &= \sum_{n=i}^{\infty} (1-p)^{n-1} p \\ &= (1-p)^{i-1} \\ \mathbb{E}[X] &= \sum_{i=1}^{\infty} \mathbb{P}(X \geq i) \\ &= \sum_{i=1}^{\infty} (1-p)^{i-1} \\ &= \frac{1}{1 - (1-p)} \\ &= \frac{1}{p} \end{aligned}$$

giving exactly the same result as the proof by the memoryless property.

Exercise: Prove that for X a continuous random variable,

$$\mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X \geq y) dy$$

Back to the Coupon Collector Problem, we have

$$\mathbb{E}[X_i] = \frac{1}{p_i} = \frac{1}{1 - \frac{i-1}{n}}$$

so

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{i=1}^n \mathbb{E}[X_i] \\
&= \sum_{i=1}^n \frac{n}{n-i+1} \\
&= nH_n \\
&= n \log n + O(n)
\end{aligned}$$

where the final step follows from the fact that

$$H_n = \sum_{i=1}^n 1/i = \log n + O(1)$$

Letting $X(\alpha)$ be the number of balls placed until αn bins are filled, we have

$$\begin{aligned}
\mathbb{E}[X(\alpha)] &= \sum_{i=1}^{\alpha n} \frac{n}{n-i+1} \\
&\approx \sum_{i=(1-\alpha)n}^n \frac{n}{i} \\
&= n \left(\sum_{i=1}^n \frac{1}{i} - \sum_{i=1}^{(1-\alpha)n} \frac{1}{i} \right) \\
&= n \log \frac{1}{1-\alpha}
\end{aligned}$$

In particular, this means that as long as we only want to fill a constant fraction of the bins, we only need to place $O(n)$ balls. If we want to fill all the bins, however, we need to place $n \log n$ balls.

Motivation: Suppose you have two job offers from a big company and a startup with stock options. The big company has a salary of \$100k and the startup has a salary of \$50k plus with stock options that will be worth \$5M with probability 1/100. Which should you choose?

Under the expected value criterion, the two offers are equivalent. However, the startup has a much higher *risk*. How might we quantify this risk?

Markov Inequality: For any non-negative random variable X and any $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Proof: Intuitively, if the expected value of X is small, then the mass of the distribution must be concentrated on small values.

$$\mathbb{E}[X] = \sum i \mathbb{P}(X = i) \geq a \sum_{i \geq a} \mathbb{P}(X = i) = a \mathbb{P}(X \geq a)$$

In the coupon collector problem, we found $\mathbb{E}[X] = nH_n = n \log n + O(n)$. What is the probability that we are very unlucky? By Markov's inequality,

$$\mathbb{P}(X \geq 2\mathbb{E}[X]) \leq \mathbb{P}(X \geq 2n \log n) \leq \frac{1}{2}$$

This is helpful but we can do better. This will also motivate another essential theme for this class: *with more information, we can make tighter bounds.*

Variance: $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

Intuitively, we want the expectation of the deviation

$$\sum (i - \mathbb{E}[X])p_i(x)$$

but of course, this is always zero. Hence, we square the deviation to get a non-negative quantity and to avoid working with an absolute value. As we will see, choosing the square also allows the very useful application of moments.

Exercise: Calculate the variance of the following RVs.

1. $X \sim \text{Bernoulli}(1/2)$

$$\text{Var } X = \frac{1}{2}(1 - \frac{1}{2})^2 + \frac{1}{2}(0 - \frac{1}{2})^2 = \frac{1}{4}$$

2. $X \sim \text{Unif}(n)$

$$\mathbb{E}[X] = \frac{1}{n} \sum_{i=1}^n i = \frac{n+1}{2}$$

$$\mathbb{E}[X^2] = \frac{1}{n} \sum_{i=1}^n i^2 = \frac{(n+1)(2n+1)}{6}$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{n^2-1}{12}$$

Notice that the variance is always non-negative so in particular, $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$.

Standard Deviation: $\sigma(X) = \sqrt{\text{Var}(X)}$

Chebyshev's Inequality: For any random variable X ,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

Proof:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) = \mathbb{P}((X - \mathbb{E}[X])^2 \geq a^2)$$

Now our RV $(X - \mathbb{E}[X])^2$ is non-negative so we can apply Markov's inequality to get

$$\mathbb{P}((X - \mathbb{E}[X])^2 \geq a^2) \leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{a^2} = \frac{\text{Var}(X)}{a^2}$$

Corollaries: For any random variables,

- 1.

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a\sigma(X)) \leq \frac{1}{a^2}$$

- 2.

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \varepsilon \mathbb{E}[X]) \leq \frac{\text{Var}[X]}{\varepsilon^2 (\mathbb{E}[X])^2}$$

Theorem: For X, Y independent random variables,

$$\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

Proof:

$$\begin{aligned}\mathbb{E}[XY] &= \sum_i \sum_j ij \mathbb{P}(X = i \wedge Y = j) \\ &= \sum_i \sum_j ij \mathbb{P}(X = i) \mathbb{P}(Y = j) \\ &= \left(\sum_i p_i \mathbb{P}(X = i) \right) \left(\sum_j \mathbb{P}(Y = j) \right) \\ &= \mathbb{E}[X] \cdot \mathbb{E}[Y]\end{aligned}$$

Theorem: If X, Y are independent random variables, then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

Proof:

$$\begin{aligned}\text{Var}[X + Y] &= \mathbb{E}[(X + Y - \mathbb{E}[X] - \mathbb{E}[Y])^2] \\ &= \mathbb{E}[(X - \mathbb{E}[X])^2 + (Y - \mathbb{E}[Y])^2 + 2(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= \text{Var}[X] + \text{Var}[Y] + 2\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= \text{Var}[X] + \text{Var}[Y] + 2\mathbb{E}[X - \mathbb{E}[X]] \cdot \mathbb{E}[Y - \mathbb{E}[Y]] \\ &= \text{Var}[X] + \text{Var}[Y]\end{aligned}$$

since $X - \mathbb{E}[X]$ and $Y - \mathbb{E}[Y]$ are independent and $\mathbb{E}[X - \mathbb{E}[X]] = \mathbb{E}[X] - \mathbb{E}[X] = 0$

Exercise: Calculate the variance of $X \sim G(p)$

Let $Y = 1$ if the first trial is a success and $Y = 0$ otherwise. Then

$$\begin{aligned}\mathbb{E}[X^2] &= \mathbb{P}(Y = 0)\mathbb{E}[X^2 \mid Y = 0] + \mathbb{P}(Y = 1)\mathbb{E}[X^2 \mid Y = 1] \\ &= (1 - p)\mathbb{E}[X^2 \mid Y = 0] + p\mathbb{E}[X^2 \mid Y = 1]\end{aligned}$$

If $Y = 0$, let Z be the number of trials until the first success after the first trial.

$$\begin{aligned}\mathbb{E}[X^2] &= (1 - p)\mathbb{E}[(z + 1)^2] + p \\ &= (1 - p)\mathbb{E}[Z^2] + 2(1 - p)\mathbb{E}[Z] + (1 - p) + p \\ &= (1 - p)\mathbb{E}[X^2] + 2(1 - p)\frac{1}{p} + 1 \\ &= (1 - p)\mathbb{E}[X^2] + \frac{2 - p}{p} \\ &= \frac{2 - p}{p^2} \text{Var}[X] &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \frac{2 - p}{p^2} - \frac{1}{p^2} \\ &= \frac{1 - p}{p^2}\end{aligned}$$

since $\mathbb{E}[Z] = 1/p$ and $\mathbb{E}[Z^2] = \mathbb{E}[X^2]$ by the memoryless property.

And all together, this does give us a tighter bound on the probability of being very unlucky in the coupon collector problem.

$$\begin{aligned}\text{Var}[X_i] &\leq \frac{1}{p^2} \\ &\leq \left(\frac{n}{n-i+1}\right)^2 \\ \text{Var}[X] &= \sum_{i=1}^n \text{Var}[X_i] \\ &\leq \sum_{i=1}^n \left(\frac{n}{n-i+1}\right)^2 \\ &= n^2 \sum_{i=1}^n \frac{1}{i^2} \\ &\leq n^2 \frac{\pi^2}{6}\end{aligned}$$

since $\sum_{i=1}^{\infty} 1/i^2 = \pi^2/6$.

By Chebyshev's inequality,

$$\mathbb{P}(|X - nH_n| \geq nH_n) \leq \frac{n^2\pi^2}{6(nH_n)^2} = O\left(\frac{1}{\log^2 n}\right)$$

Can we do even better?

Consider the probability of not hitting the i th bin after $n \log n + cn$ trials. This is

$$\left(1 - \frac{1}{n}\right)^{n \log n + cn} < e^{-\log n + c} = \frac{1}{e^c n}$$

Union Bound: Let $E_1, \dots, E_n$ be any n events in a probability space. Then

$$\mathbb{P}\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n \mathbb{P}(E_i)$$

Hence, the probability of not hitting any bin after $n \log n + cn$ trials is at most e^{-c} .

The probability that all bins have not been hit after $2n \log n$ trials is at most $1/n$.

And finally, the probability that all bins have not been hit after $n(\log n + 2 \log \log n)$ trials is at most $O(1/\log^2 n)$.

0.6 Sept 29

A typical probability theory statement makes claims in the limit of trials. For example

Central Limit Theorem: For $X_1, \dots, X_n$ independent and identically distributed random variables with mean μ and variance σ^2 ,

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(\frac{1}{\sigma n \sqrt{n}} \sum_{i=1}^n (X_i - \mu) \leq z\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt$$

However, this tells us nothing about the speed of convergence. In probabilistic methods for CS, we are interested in *finite sample approximation* and explicit, feasible bounds on the probability of error.

0.7 Sept 29

Notice that in our analysis of the quicksort algorithm, we could only talk about the expectation of the run time. We could not say anything about the probability of the run time being much larger than the expectation. Today, we will introduce *large deviation bounds*.

Chernoff (Large Deviation) Bound: Let $X_1, \dots, X_n$ be independent Bernoulli random variables with $\mathbb{P}(X_i = 1) = \mathbb{E}[X_i] = p_i$. Let $\mu = \sum_{i=1}^n p_i$. Then for any $\delta > 0$,

1.

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i \geq (1 + \delta)\mu\right) \leq e^{-n\mu\delta^2/3}$$

2.

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i \leq (1 - \delta)\mu\right) \leq e^{-n\mu\delta^2/2}$$

That is, the probability of a large deviation from the mean is exponentially small in n .

Example: Consider n coin flips. Let X be the number of heads.

- Markov gives

$$\mathbb{P}(X \geq \frac{3n}{4}) \leq \frac{n/2}{3n/4} \leq \frac{2}{3}$$

- Chebyshev gives

$$\mathbb{P}\left(\left|X - \frac{n}{2}\right| \geq \frac{n}{4}\right) \leq \frac{4}{n}$$

- Chernoff gives

$$\mathbb{P}\left(\left|X - \frac{n}{2}\right| \geq \frac{n}{4}\right) \leq \mathbb{P}\left(X \geq \frac{n}{2}\left(1 + \frac{1}{2}\right)\right) + \mathbb{P}\left(X \leq \frac{n}{2}\left(1 - \frac{1}{2}\right)\right) \leq 2e^{-n/24}$$

Notice that each of these bounds is successively tighter but uses more information about the distribution of X (just the expectation, the variance, and everything).

The basic idea of the proof of the Chernoff bound is to use Markov's inequality on the moment generating function of X : for any $t > 0$

$$\mathbb{P}(X \geq a) = \mathbb{P}(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}}$$

and similarly for $t < 0$:

$$\mathbb{P}(X \leq a) = \mathbb{P}(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}}$$

Of course, the real challenge is to compute $\mathbb{E}[e^{tX}]$.

Theorem: Let $X_1, \dots, X_n$ be independent Bernoulli random variables with $\mathbb{P}(X_i = 1) = p_i$. Let $X = \sum_{i=1}^n X_i$ and

$$\mu = \mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X_i] = \sum_{i=1}^n p_i$$

Then

1. For any $\delta > 0$,

$$\mathbb{P}(X \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{(1 + \delta)}} \right)^\mu$$

2. For $0 < \delta < 1$,

$$\begin{aligned} \mathbb{P}(X \geq (1 + \delta)\mu) &\leq e^{-\mu\delta^2/3} \\ \mathbb{P}(X \leq (1 - \delta)\mu) &\leq \left(\frac{e^{-\delta}}{(1 - \delta)^{(1 - \delta)}} \right)^\mu \leq e^{-\mu\delta^2/2} \end{aligned}$$

3. For $R > 6\mu$,

$$\mathbb{P}(X \geq R) \leq 2^{-R}$$

Proof: For each X_i ,

$$\begin{aligned} M_{X_i}(t) &= \mathbb{E}[e^{tX_i}] \\ &= p_i e^t + (1 - p_i) \\ &= 1 + p_i(e^t - 1) \\ &\leq e^{p_i(e^t - 1)} \end{aligned}$$

since $X_i \in \{0, 1\}$ and $1 + x \leq e^x$ for all x .

Notice that this is a function of t since we took the expectation over X_i .

Now we can compute the moment generating function of X :

$$\begin{aligned} M_X(t) &= \mathbb{E}[e^{tX}] \\ &= \mathbb{E}\left[\exp\left(t \sum_{i=1}^n X_i\right)\right] = \mathbb{E}\left[\prod_{i=1}^n e^{tX_i}\right] = \prod_{i=1}^n \mathbb{E}[e^{tX_i}] \\ &\leq \prod_{i=1}^n \exp(p_i(e^t - 1)) \\ &= \exp\left((e^t - 1) \sum_{i=1}^n p_i\right) = e^{(e^t - 1)\mu} \end{aligned}$$

Applying Markov's inequality for $t > 0$ gives

$$\begin{aligned} \mathbb{P}(X \geq (1 + \delta)\mu) &= \mathbb{P}(e^{tX} \geq e^{t(1 + \delta)\mu}) \\ &\leq \frac{\mathbb{E}[e^{tX}]}{e^{t(1 + \delta)\mu}} \\ &\leq \frac{e^{(e^t - 1)\mu}}{e^{t(1 + \delta)\mu}} \end{aligned}$$

For any $\delta > 0$, we can set $t = \log(1 + \delta) > 0$ to get

$$\mathbb{P}(X \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{(1 + \delta)}} \right)^\mu$$

For $\delta > 0$, we can set $t = \log(1 + \delta) > 0$ to get

$$\mathbb{P}(X \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \right)^\mu$$

This proves (1).

For (2), we first show

$$\frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \leq e^{-\delta^2/3}$$

or equivalently, that

$$f(\delta) = \delta - (1 + \delta) \log(1 + \delta) + \delta^2/3 \leq 0$$

with $\delta \in (0, 1)$.

Since $f''(\delta) < 0$ for $\delta \in [0, 1/2)$ and $f''(\delta) > 0$ for $\delta \in [1/2, 1)$, we know $f'(\delta)$ first decreases then increases over $[0, 1]$.

Since $f'(0) = 0$ and $f'(1) < 0$, we have that $f'(\delta) \leq 0$. Finally, $f(0) = 0$ and f is decreasing so $f(\delta) \leq 0$ in the interval, giving

$$\boxed{\mathbb{P}(X \geq (1 + \delta)\mu) \leq e^{-\mu\delta^2/3}}$$

On the other hand, using Markov for all $t < 0$, we have that

$$\mathbb{P}(X \leq (1 - \delta)\mu) = \frac{e^{(e^t - 1)\mu}}{e^{t(1 - \delta)\mu}}$$

by the same argument above.

On $0 < \delta < 1$, set $t = \log(1 - \delta) < 0$ to get

$$\mathbb{P}(X \leq (1 - \delta)\mu) \leq \left(\frac{e^{-\delta}}{(1 - \delta)^{(1-\delta)}} \right)^\mu$$

Then by similar argument as above, we can show

$$f(\delta) = -\delta - (1 - \delta) \log(1 - \delta) + \frac{1}{2}\delta^2 \leq 0$$

to get

$$\boxed{\mathbb{P}(X \leq (1 - \delta)\mu) \leq e^{-\mu\delta^2/2}}$$

For (3), with $R \geq 6\mu$ and $\delta \geq 5$,

$$\mathbb{P}(X \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \right)^\mu \leq \left(\frac{e}{6} \right)^R \leq 2^{-R}$$

This completes the proof.

The coin flipping example above asks: how likely is it that we will see $3/4$ heads in n flips of a fair coin? In fact,

we can get a much tighter bound using Chernoff:

$$\begin{aligned}\mathbb{P}\left(\left|X - \frac{n}{2}\right| \geq \frac{1}{2}\sqrt{6n \log n}\right) &= \mathbb{P}\left(X \geq \frac{n}{2} + \frac{1}{2}\sqrt{\frac{6 \log n}{n}}\right) + \mathbb{P}\left(X \leq \frac{n}{2} - \frac{1}{2}\sqrt{\frac{6 \log n}{n}}\right) \\ &\leq \exp\left(-\frac{1}{3} \frac{n}{2} \frac{6 \log n}{n}\right) + \exp\left(-\frac{1}{2} \frac{n}{2} \frac{6 \log n}{n}\right) \\ &\leq \frac{2}{n}\end{aligned}$$

using the fact that $\sigma = \sqrt{n/4}$ and choosing our constants carefully.

The obvious question is: why are we choosing a deviation of $\sqrt{n \log n}$?

Simply, $\sqrt{n}$ is the order of the standard deviation of X so we could not hope to get a bound on the probability of a deviation of order $\sqrt{n}$ that is better than $O(1)$. The $\log n$ factor is a well-chosen trick to get a bound that goes to zero as $n \rightarrow \infty$.

Theorem: Let $X_1, \dots, X_n$ be independent Bernoulli random variables with $\mathbb{P}(X_i = 1) = p_i$. Let $X = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[X]$ then define $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\bar{\mu} = \mathbb{E}[\bar{X}]$.

We showed above that

$$\begin{aligned}\mathbb{P}(X \geq (1 + \delta)\mu) &\leq e^{-\mu\delta^2/3} \\ \mathbb{P}(X \leq (1 - \delta)\mu) &\leq e^{-\mu\delta^2/2}\end{aligned}$$

But additionally,

$$\begin{aligned}\mathbb{P}(\bar{X} \geq (1 + \delta)\bar{\mu}) &\leq e^{-\bar{\mu}\delta^2/3} \\ \mathbb{P}(\bar{X} \leq (1 - \delta)\bar{\mu}) &\leq e^{-\bar{\mu}\delta^2/2}\end{aligned}$$

Notice the expectation is in the exponent. If, however, we have an RV with 0 mean, this is useless! We will see next time a different Chernoff bound that does not depend on the expectation of X .

Example (Monte Carlo Methods): Estimate the value of π .

Choose X, Y independently and uniformly at random in $[0, 1]$. Let

$$Z = \begin{cases} 1 & \sqrt{X^2 + Y^2} \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Then $\frac{1}{2} \leq p = \mathbb{P}(Z = 1) = \frac{\pi}{4} \leq 1$ and $4\mathbb{E}[Z] = \pi$.

Let $Z_1, \dots, Z_n$ be the values of Z for m independent experiments. Define $W_m = \sum_{i=1}^m Z_i$.

$$\mathbb{E}[W_m] = \sum_{i=1}^m \mathbb{E}[Z_i] = m\mathbb{E}[Z] = \frac{m\pi}{4}$$

Then $\tilde{\pi}_m = \frac{4}{m}W_m$ is an *unbiased estimator* of π , that is $\mathbb{E}[\tilde{\pi}_m] = \pi$.

How many samples do we need to get a good estimate of π ?

$$\begin{aligned}
 \mathbb{P}(|\tilde{\pi}_m - \pi| \geq \varepsilon\pi) &= \mathbb{P}\left(\left|W - \frac{m\pi}{4}\right| \geq \frac{m\varepsilon\pi}{4}\right) \\
 &= \mathbb{P}(|W_m - \mathbb{E}[W_m]| \geq \varepsilon\mathbb{E}[W_m]) \\
 &= \mathbb{P}(W_m - \mathbb{E}[W_m] \geq \varepsilon\mathbb{E}[W_m]) + \mathbb{P}(W_m - \mathbb{E}[W_m] \leq -\varepsilon\mathbb{E}[W_m]) \\
 &\leq \exp(-\frac{1}{3}\frac{m\pi}{4}\varepsilon^2) + \exp(-\frac{1}{2}\frac{m\pi}{4}\varepsilon^2) \\
 &\leq 2\exp(-\frac{1}{12}m\pi\varepsilon^2)
 \end{aligned}$$

But this is a problem! We don't know the value of π in this expression!

However, we can go back to geometry to bound $\pi > 2$ and get

$$\mathbb{P}(|\tilde{\pi}_m - \pi| \geq \varepsilon\pi) \leq 2\exp(-\frac{1}{12}m\pi\varepsilon^2) \leq \exp(-\frac{1}{6}m\varepsilon^2) = \delta$$

So

$$6m\varepsilon^2 \geq \log \frac{1}{\delta} \implies m \geq \frac{6\log(1/\delta)}{\varepsilon^2} + o(1)$$

but this is not ideal since $\varepsilon \ll 1$ will make m very large.

For example, if we want an error $\leq 1\%$ ($\varepsilon = 0.1$ and $\delta = 0.01$), we need $m \geq 4000$.

Let's return to the general scheme. For any random variable X , we can calculate large deviations bounds via

1. computing $\mathbb{E}[e^{tX}]$
2. optimize

$$\begin{aligned}
 \mathbb{P}(X \geq a) &\leq \min_{t>0} \frac{\mathbb{E}[e^{tX}]}{e^{ta}} \\
 \mathbb{P}(X \leq a) &\leq \min_{t<0} \frac{\mathbb{E}[e^{tX}]}{e^{ta}}
 \end{aligned}$$

3. simplify the resulting expression to get a nice bound

It is worth exploring the moment generating function of X in more detail.

Moment Generating Function: For a random variable X , the moment generating function is defined for $t \in \mathbb{R}$ as

$$M_X(t) = \mathbb{E}[e^{tX}]$$

(though this depends on the expectation converging).

Theorem: Let X be a random variable with moment generating function $M_X(t)$. Assuming that exchanging the expectation and differentiation is valid, then for all $n \geq 1$,

$$\mathbb{E}[X^n] = M_X^{(n)}(0)$$

where $M_X^{(n)}(0)$ is the n th derivative of $M_X(t)$ evaluated at $t = 0$.

Proof:

$$M_X^{(n)}(t) = \mathbb{E}[X^n e^{tX}]$$

so

$$M_X^{(n)}(0) = \mathbb{E}[X^n]$$

Notice in particular

- $M'_x(t) = \mathbb{E}[Xe^{tX}]$ so $\mathbb{E}[X] = M'_X(0)$
- $M''_X(t) = \mathbb{E}[X^2e^{tX}]$ so $\mathbb{E}[X^2] = M''_X(0)$

which easily gives us the expectation and variance.

When can we switch the order of differentiation and expectation? This is largely out of scope but we can notice that

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_{i \in \text{dom}(X)} e^{tx_i} \cdot \mathbb{P}(X = i)$$

Assuming the sum is finite or converges uniformly, we can differentiate term by term to get

$$M'_X(t) = \sum_{i \in \text{dom}(X)} \frac{d}{dt} e^{ti} \mathbb{P}(X = i) = \mathbb{E}\left[\frac{d}{dt} e^{ti}\right]$$

Theorem: Let X, Y be independent random variables with moment generating functions $M_X(t), M_Y(t)$. Then the moment generating function of $X + Y$ is

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$$

Proof:

$$M_{X+Y}(t) = \mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX}] \cdot \mathbb{E}[e^{tY}] = M_X(t) \cdot M_Y(t)$$

Theorem: Let X, Y be two random variables. If

$$M_X(t) = M_Y(t)$$

for all $t \in (-\delta, \delta)$ for some $\delta > 0$, then X and Y have the same distribution.

Proof: Out of scope.